Do Momentum Signals Beat Mean Reversion in Crypto? Evidence from Binance Spot Markets

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1. Introduction

Crypto assets exhibit high volatility, fragmented liquidity, and return drivers that differ from those of conventional equity and macro factors (Liu & Tsyvinski, 2021). These features make them a natural environment for systematic trading, but they also make turnover, leverage, and execution frictions first-order design constraints. This report studies a liquid Binance USDT universe consisting of BTC, ETH, BNB, XRP, ADA, and DOGE sampled at the hourly frequency from 1 January 2022 to 15 March 2026. The starting capital is $10,000 USDT and the gross exposure constraint is

$$\sum_i |\theta_t^i| \leq 100{,}000,$$

so the maximum book size is ten times initial capital.

Two strategies are implemented. The first is a time-series trend-following rule motivated by medium-horizon return continuation and technical momentum (Brock et al., 1992; Moskowitz et al., 2012). The second is a short-horizon cross-sectional mean-reversion strategy motivated by relative-value and contrarian effects (Gatev et al., 2006; Avellaneda & Lee, 2010). In both cases the raw signal is passed through the same cost-aware portfolio construction layer so that performance differences are due primarily to the underlying forecasting relation rather than to inconsistent sizing rules.

The central question is which signal family, if any, remains profitable after realistic slippage once positions are expressed in USDT notionals and constrained by the $100k gross cap. The empirical objectives are to: construct and clean a multi-asset hourly crypto panel; form excess returns and assess whether the risk-free leg matters at this horizon; estimate a one-way slippage parameter compatible with the coursework PnL definition; and compare gross and net performance, turnover, holding horizon, and downside risk across the two strategies.

2. Data

The asset universe contains BTCUSDT, ETHUSDT, BNBUSDT, XRPUSDT, ADAUSDT, and DOGEUSDT. These names were selected because they are among the most liquid Binance spot pairs and provide a useful mix of market beta and altcoin-specific dispersion, which is desirable for both momentum and relative-value signals. The data consist of Binance spot OHLCV bars at the 1-hour frequency. The final sample runs from 2022-01-01 00:00 UTC to 2026-03-15 15:00 UTC and contains 36,832 hourly observations for each asset, i.e. a balanced panel of 220,992 bars.

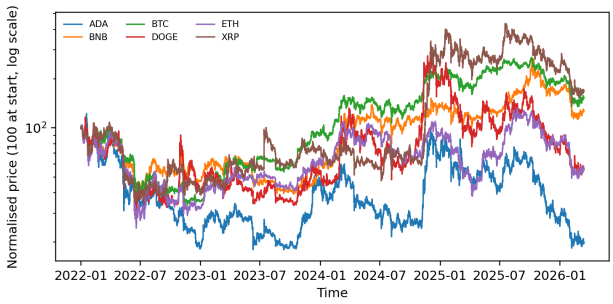

Figure 1. Normalised hourly closes (100 at the start, log scale). The universe spans two major market regimes and substantial cross-sectional divergence, motivating both time-series and relative-value signals.

Cleaning is deliberately conservative because large jumps are common in crypto and should not be removed merely for being extreme. First, each symbol is reindexed to a complete hourly grid. The sample contains only one missing bar per asset; these gaps are filled by carrying forward the close, setting open/high/low equal to the filled close, and setting volume and trade counts to zero. Second, a local "spike-and-reversal" filter is applied to log-closes. A bar is repaired only if it is bracketed by two large opposite-signed jumps while the two-bar move is small, which is consistent with a transient bad tick rather than a genuine price move. In the realised sample no bar triggered this repair rule, so the cleaned panel is effectively the raw panel plus gap filling.

Let $p_{i,t}$ denote the cleaned close. Excess simple returns are formed as

$$r_{i,t}^e = \frac{p_{i,t} - p_{i,t-1}}{p_{i,t-1}} - r_{t-1}^f,$$

where r_{t-1}^f is the lagged risk-free rate aligned to the hourly bar. We use the FRED DFF series and convert the daily annualised rate into an hourly simple rate by linear day-count scaling. The risk-free adjustment is economically negligible at this frequency: the average absolute hourly crypto return is 50.31 bps, while the average lagged hourly risk-free rate is only 0.0457 bps, or 0.09% of the average absolute bar move. Hence the excess-return transformation is conceptually correct but has almost no effect on trading decisions or PnL at the hourly horizon.

Figure 1 shows the normalised price paths. The universe exhibits clear regime variation, large cross-sectional dispersion, and volatility clustering, all of which are characteristic

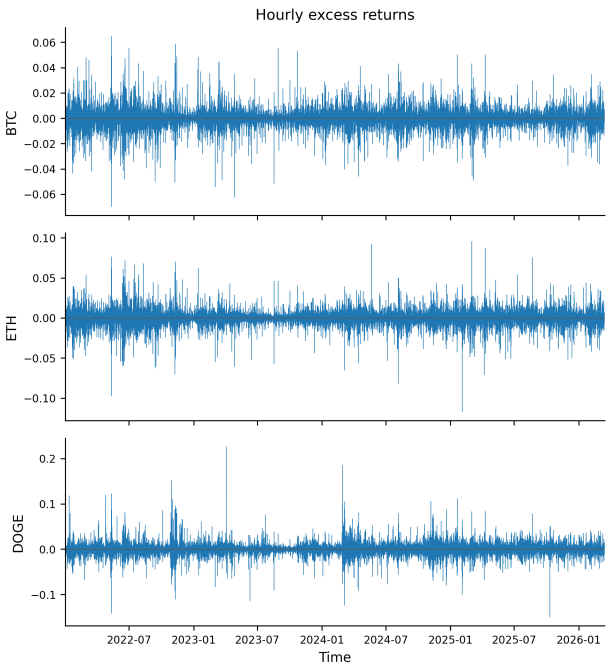

Figure 2. Representative hourly excess returns. DOGE is visibly more heteroskedastic and heavy-tailed than BTC and ETH, while all three series show clustered volatility and occasional jump episodes.

of financial return series and consistent with the broader evidence on conditional heteroskedasticity. BTC and XRP dominate the late-sample rally, while ADA and ETH are weaker on a normalised basis, which creates scope both for directional trend capture and for relative-value spreads.

For reproducibility, both the raw Binance OHLCV download and the cleaned balanced panel are stored in `parquet` format prior to portfolio construction. In addition to price levels, the hourly excess return series are plotted directly, since the project brief requires analysis of returns rather than only normalised prices. The return panel highlights the main empirical features of the sample: volatility clustering, episodic jumps, and substantial cross-sectional heterogeneity across assets. These features are important for the later strategy design, because they create scope for both persistence-based signals and relative-value dislocations, while also showing why outlier treatment must remain conservative in crypto markets.

3. Trading strategy design

Both strategies share the same portfolio construction layer. Denote by w_t the vector of portfolio weights on the \$100k gross-cap budget, so that $\theta_t = 100{,}000\, w_t$. At each hour we solve the cost-aware quadratic program

$$\max_{w_t} \; \mu_t^\top w_t - \frac{\lambda}{2} w_t^\top \widehat{\Sigma}_t w_t - \frac{\kappa}{2}\|w_t - w_{t-1}\|_2^2 \quad \text{s.t.} \quad \|w_t\|_1 \le 1,$$

where μ_t is the signal-implied expected return vector, $\widehat{\Sigma}_t$ is a rolling covariance estimate, λ is risk aversion, and κ penalises turnover. This is a practical approximation to dynamic mean-variance trading with transaction costs

Table 1. Selected strategy settings from the Part 2 parameter search. Validation and test Sharpe ratios are *gross* and correspond to the pre-cost strategy layer.

Strategy	L (h)	κ	Val. Sharpe	Test Sharpe	Val. turn. (kUSDT/bar)
Trend	336	0.02	2.52	0.34	2.53
Mean rev.	24	0.10	-1.21	0.62	7.46

Table 2. Asset-level one-way slippage estimates. The selected scalar used in PnL attribution is the turnover-weighted average across both strategies, $\hat{s} = 8.68$ bps.

Asset	Method	s_i (bps)	Avg. quote vol. (m USDT/hr)
BNB	Roll	5.23	9.0
BTC	Roll	5.97	104.9
ETH	Corwin-Schultz	6.92	51.3
ADA	Roll	9.06	4.1
XRP	Roll	10.41	12.9
DOGE	Roll	11.58	9.6

(Markowitz, 1952; Garleanu & Pedersen, 2013). The covariance matrix uses a 168-hour rolling window with 25% diagonal shrinkage for stability in the spirit of **?**. The risk-aversion parameter is fixed at $\lambda = 15$ for both strategies, while signal-specific lookback windows and turnover penalties are selected on a 60/20/20 train/validation/test split after the warm-up period.

The important distinction is that the brief is not asking only for two forecasting signals; it is asking for two complete trading strategies, meaning a signal layer, a sizing rule, and an explicit mechanism for enforcing the leverage limit at each rebalancing time. In our setup, each strategy first produces a vector of signal-implied expected returns μ_t, and the portfolio construction layer then maps these signals into feasible notionals θ_t on the common \$100k gross-exposure budget.

The brief also requires discussion of parameter choice and how those choices were made. For that reason, the strategy design below is organised around the same sequence for both models: economic intuition, signal definition, trading interpretation, and parameter selection. Parameters are not chosen ad hoc from the full sample; instead, they are selected from small pre-specified grids using a train/validation/test split, and then locked before the final full-sample run.

3.1. Strategy 1: Trend-Following

The economic idea is return continuation driven by delayed information diffusion, behavioural underreaction, and technical breakout dynamics (Brock et al., 1992; Moskowitz et al., 2012). For asset i and lookback L, define the trend score

$$m_{i,t}^{\mathrm{TR}}(L) = \log\left(\frac{P_{i,t}}{P_{i,t-L}}\right), \qquad z_{i,t}^{\mathrm{TR}}(L) = \frac{m_{i,t}^{\mathrm{TR}}(L)}{\hat{\sigma}_{i,t}\sqrt{L}},$$

where $\hat{\sigma}_{i,t}$ is the lagged 1-bar volatility estimate from a 168-hour rolling standard deviation. The raw signal is $\tanh(z_{i,t}^{\mathrm{TR}})$, shifted by one bar to avoid look-ahead bias, and the expected-return input is $\mu_{i,t} = \hat{\sigma}_{i,t}\tanh(z_{i,t-1}^{\mathrm{TR}})$. The hy-

perbolic tangent truncates extreme scores and prevents the allocator from concentrating the full book on a single asset.

There is no hard entry threshold. A long position is entered when the optimal weight becomes positive, a short when it becomes negative, and positions are partially or fully exited whenever the optimizer moves the target back toward zero or reverses sign. Candidate lookbacks are $L \in \{24, 72, 168, 336\}$ hours and $\kappa \in \{0.02, 0.05, 0.10\}$. Validation selects $L = 336$ and $\kappa = 0.02$, delivering a gross validation Sharpe of 2.52 and a gross test Sharpe of 0.34. The selected model therefore favours a relatively slow signal and a light turnover penalty, consistent with the idea that continuation in crypto is stronger at multi-day rather than intraday horizons.

3.2. Strategy 2: Mean-Reversion

The second strategy is a short-horizon relative-value reversal rule. This is closer in spirit to contrarian and statistical-arbitrage trading than to cointegration-based pairs formation, but the same broad logic applies: temporary cross-sectional dislocations should mean-revert if recent winners have become locally overextended and recent losers oversold (Engle & Granger, 1987; Gatev et al., 2006; Avellaneda & Lee, 2010). For lookback L, define the cross-sectionally demeaned short-horizon move

$$m_{i,t}^{\mathrm{MR}}(L) = \log\left(\frac{P_{i,t}}{P_{i,t-L}}\right) - \frac{1}{N_t}\sum_{j=1}^{N_t}\log\left(\frac{P_{j,t}}{P_{j,t-L}}\right),$$

with standardised score $z_{i,t}^{\mathrm{MR}} = m_{i,t}^{\mathrm{MR}}/(\hat{\sigma}_{i,t}\sqrt{L})$. The signal is the negative of the lagged bounded score, $-\tanh(z_{i,t-1}^{\mathrm{MR}})$, so recent relative winners are sold and losers bought. To keep the book approximately market-neutral, the expected-return vector is demeaned across active assets before solving the optimizer.

Candidate lookbacks are $L \in \{4, 8, 12, 24\}$ hours with the same turnover-penalty grid. Validation selects $L = 24$ and $\kappa = 0.10$. Crucially, every mean-reversion candidate has negative validation Sharpe, so the retained specification should be interpreted as the least-poor member of a weak strategy family rather than as a strongly supported alpha source. The higher selected turnover penalty is economically sensible because short-horizon reversal signals refresh rapidly and are more exposed to microstructure noise.

4. Transaction costs and slippage model

Slippage is estimated as a one-way decimal trading cost compatible with the coursework cost function

$$\mathrm{Cost}_t = s\sum_i \left|\theta_t^i - \theta_{t-1}^i(1 + r_{i,t-1})\right|.$$

This section addresses the specific transaction-cost requirement of the brief by estimating a slippage parameter s that is directly compatible with the portfolio cost term above. The objective is not to recover the full fee schedule or exact

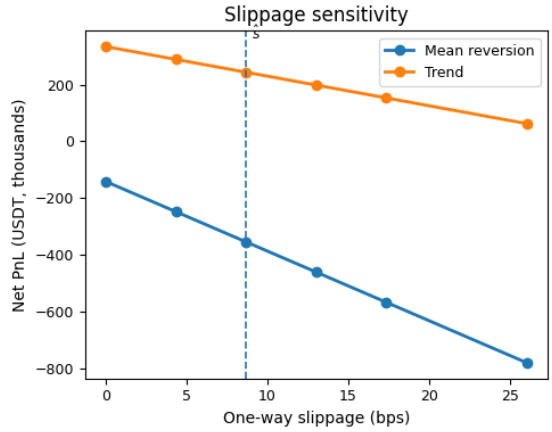

Figure 3. Sensitivity of net PnL to one-way slippage. The dashed line marks the turnover-weighted estimate $\hat{s} = 8.68$ bps. Trend-following remains positive over the tested range, while mean-reversion is negative even at zero cost.

realised execution shortfall, but to construct a defensible one-way trading-cost proxy, expressed as a decimal per trade, that can be applied consistently across assets and strategies. Accordingly, we first estimate asset-level slippage from observed price dynamics, then aggregate these into a single scalar $\hat{s}$ for portfolio-level PnL attribution, and finally study how the net results change when the assumed slippage level is varied. This matches the two core questions posed by the assignment: how to estimate s, and how sensitive the strategies are to that estimate.

The primary estimator is the Roll model, which links effective bid-ask spread to the negative first-order covariance of price changes. For asset i we compute a rolling 30-day (720-hour) estimator,

$$\hat{s}_{i,t}^{\mathrm{Roll}} = \frac{\sqrt{\max\{-\widehat{\mathrm{Cov}}(\Delta p_{i,t}, \Delta p_{i,t-1}), 0\}}}{\widetilde{p}_{i,t}},$$

where $\widetilde{p}_{i,t}$ is a rolling median price used to convert the implied half-spread from price units into a proportional one-way cost. When the rolling Roll covariance is non-negative for extended periods, the estimator collapses to zero; ETH is the main example. In that case we use the Corwin-Schultz high-low estimator as a fallback (Corwin & Schultz, 2012).

The selected one-way slippage estimates are economically plausible and increase as liquidity falls: 5.23 bps for BNB, 5.97 for BTC, 6.92 for ETH, 9.06 for ADA, 10.41 for XRP, and 11.58 for DOGE. The turnover-weighted scalar used in performance attribution is $\hat{s} = 8.68$ bps one-way. Figure 3 shows the sensitivity of net PnL to alternative slippage assumptions. The relationship is approximately linear because costs are proportional to turnover. At $\hat{s}$, slippage consumes 27.1% of trend-following gross PnL but 149.9% of mean-reversion gross PnL, immediately revealing that turnover is the dominant weakness of the reversal strategy. The ranking is robust over a wide range of assumed costs.

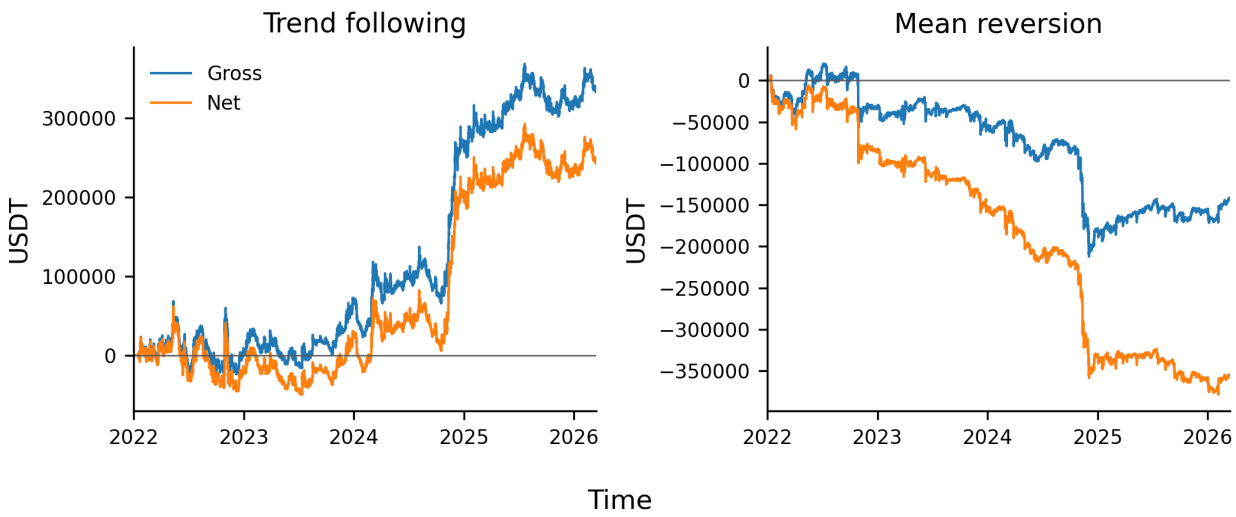

Figure 4. Gross versus net cumulative PnL under the fixed $100k gross-cap rule. Transaction costs materially reduce trend profits and decisively worsen the already negative mean-reversion sleeve.

5. Performance evaluation

Table 3 reports the primary results under the coursework's *fixed* $100k gross-cap interpretation. Any capital not allocated to active positions is held as USDT cash, but profitable strategies are not allowed to lever above the fixed $100k ceiling. Because the fixed-cap rule can drive net equity below zero while trading continues, the Sharpe, Sortino, and Calmar ratios in the table are computed from hourly PnL scaled by initial capital rather than by current equity; this keeps the ratios interpretable even after insolvency events.

The economic conclusion is unambiguous. Trend-following generates $333.9k gross PnL and $243.4k net PnL after $90.4k of slippage, a 2,434% return on the initial $10k account under the fixed-cap convention. Mean-reversion loses $141.9k gross and $354.5k net, with transaction costs alone exceeding $212k. Trend-following also trades much more slowly: total turnover is $104.2m versus $245.1m, and the average holding horizon is 2.92 days versus 1.23 days. Net risk-adjusted performance is positive for trend-following (Sharpe 0.64, Sortino 0.93, Calmar 3.73) and strongly negative for mean-reversion.

Figure 5a shows the full net-equity paths. Trend-following is roughly flat to mildly negative for much of 2022–2024, then benefits strongly from the directional market regime of late 2024 and 2025. Mean-reversion decays almost monotonically and suffers a sharp additional loss around the regime break in late 2024, consistent with a fast contrarian book being caught on the wrong side of persistent directional moves.

Figure 4 separates gross and net cumulative PnL. Trend-following benefits from a strong directional regime in late 2024–2025, while mean-reversion experiences persistent drawdown and a sharp deterioration during the same period. This is exactly the pattern one would expect if the market entered a strong trending regime in which short-horizon contrarian bets were repeatedly run over. A practical caveat remains essential: under the fixed-$100k rule both strategies continue trading after net equity becomes non-positive, so the reported fixed-cap results should be interpreted as the outcome of the coursework accounting convention rather than as the live path of a self-financing $10k account.

A crucial caveat is financing realism. Under the fixed-cap convention, both strategies are allowed to keep trading after net equity becomes non-positive; mean-reversion crosses zero on 14 January 2022 and trend-following on 20 March 2022. Figure 5b therefore compares trend-following against an "equity-protected" alternative that scales the gross limit with lagged net equity. Under that conservative rule the trend strategy is effectively liquidated by May 2022 and never recovers, while mean-reversion also goes to zero. Hence the impressive fixed-cap trend PnL should be read as the answer to the coursework's accounting rule, not as a realistic live track record for a stand-alone $10k account without external financing. This is arguably the most important practical lesson of the exercise.

6. Next steps, limitations, and live trading discussion

The results suggest that medium-horizon directional continuation is the only strategy family with a credible edge in this dataset, and even that edge is fragile once financing realism is imposed. Trend-following should work best in persistent, high-participation regimes with broad directional moves and moderate trading frictions; it should work poorly in choppy markets with frequent reversals and low signal-to-noise. By contrast, short-horizon mean-reversion requires a stronger microstructure or residual-reversion effect than is present here, and its turnover burden is far too large for hourly execution at the estimated spread levels.

Table 3. Primary performance results under the fixed $100k gross-cap rule. Ratios are computed from hourly PnL normalised by initial capital because current-equity returns become ill-posed once net equity turns non-positive.

Strategy	Gross (kUSDT)	Net (kUSDT)	Cost (kUSDT)	Return (%)	Turnover (mUSDT)	Hold (days)	Sharpe net	Sortino net	Calmar net
Trend-following	333.9	243.4	90.4	2434.3	104.2	2.92	0.64	0.93	3.73
Mean-reversion	-141.9	-354.5	212.7	-3545.1	245.1	1.23	-1.53	-2.06	-0.34

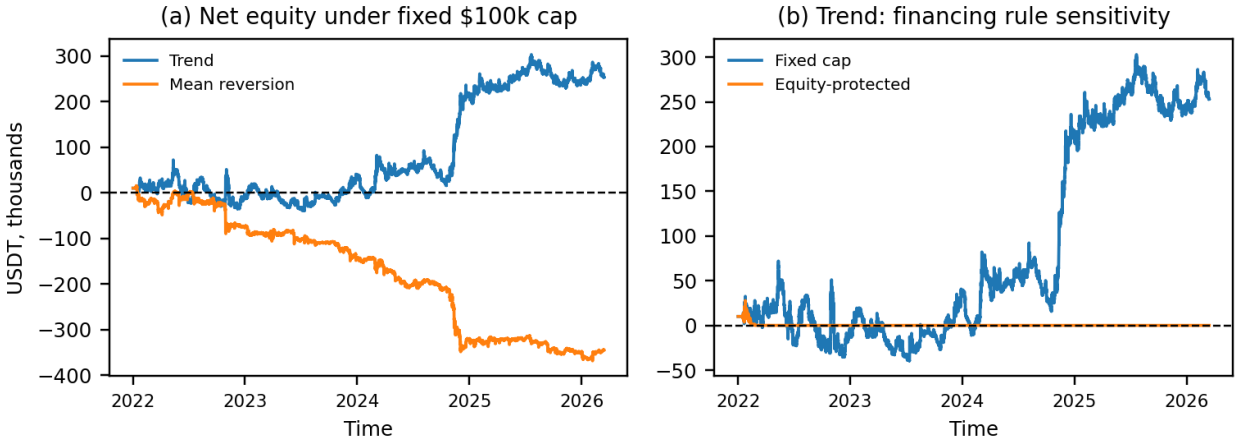

Figure 5. Performance diagnostics. Panel (a) compares net equity across strategies under the fixed $100k gross-cap rule. Panel (b) shows that the apparent success of trend-following depends critically on financing assumptions: once the gross limit is linked to current equity, the strategy is effectively liquidated near the start of the sample.

Several limitations matter for live deployment. First, the signals are estimated from spot prices but the portfolio can take signed notionals, so a real implementation would require perpetual futures or margin rather than pure spot trading. Second, the cost model includes slippage but not exchange fees, funding, borrow charges, or nonlinear market impact. Third, the fixed $100k gross cap is much too aggressive for a $10k account if capital protection is a real constraint. Finally, the strategy is regime dependent; the late-sample success of trend-following is tied to a directional market phase and may not persist.

The first is to replace the fixed-cap accounting rule with an equity-linked leverage schedule. The second is to enrich the signal set with regime filters based on realised volatility, volume, and market breadth, which may help switch between trend and reversal states. The third is to revisit the relative-value sleeve using explicit pair selection or cointegration tests rather than broad cross-sectional reversal (Engle & Granger, 1987; Gatev et al., 2006). Fourth, the cost model should be made asset- and state-dependent, for example by interacting spreads with volatility and quote volume. Finally, robustness should be re-evaluated at 4-hour and daily horizons, where reversal signals may turn over more slowly and where financing stress is less extreme.

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